

Solution Design

YAT Accounting System Cloud Architecture — Baseline Design

Engagement	YAT Accounting System Cloud Migration — Foundation Build
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1. Purpose and Scope

This document specifies the baseline AWS architecture MTS will implement as the foundation build phase of the YAT Accounting System cloud migration. It translates the approved direction into a concrete, implementable design for the Ledgerline finance and office-administration system. The design stops at “infrastructure ready for application deployment”: the EC2 instance is provisioned with the OS, the RDS instance with an empty Microsoft SQL Server engine, and the load balancer with placeholder health checks — no application binaries or financial data are placed by the MTS build.

In scope of this design

- The production cloud foundation for the Ledgerline accounting application.
- All compute, networking, identity, storage, database, autoscaling and baseline monitoring needed to run Ledgerline as a staff-facing, business-hours workload in AWS.
- Single-region, single-Availability-Zone deployment in ap-southeast-2 (Sydney).

Out of scope — deferred to the follow-on HA design phase

- High-availability hardening (Multi-AZ SQL Server, cross-AZ compute resilience, failure-simulation testing).
- Disaster recovery to a second AWS region; DR runbook and tabletop testing.
- Application re-platforming (Ledgerline remains Windows Server 2016 + Microsoft SQL Server).

Out of MTS scope entirely — YAT ICT responsibility

- Ledgerline application installation onto the EC2 instance(s) after handover.
- Database migration (extract from on-prem SQL Server, load into RDS for SQL Server).
- Cutover — DNS switch, parallel running, decommissioning, user redirection (avoiding month-end and EOFY).
- Organisational change management — CAB approvals, communications, training, post-cutover support.

2. Design Inputs and Requirements

2.1 Inputs

- Accounting System Application Specification — functions, user load, data, integrations, SLAs, data residency.
- Accounting System Migration Requirements — platform preservation, availability, recovery, licensing, retention.
- Accounting System Server Specifications — current server, measured availability, utilisation, growth.
- Engagement Role Brief and Consultation Notes — engagement scope; OS, application and database preservation.

2.2 Requirements the design must meet

Requirement	Target / note
Region / data residency	ap-southeast-2 (Sydney); financial records + staff/debtor PII within Australia (Privacy Act APP 8; financial-records retention)
Application stack	Preserved — Windows Server 2016 · Microsoft SQL Server (Standard) · Ledgerline

Concurrent users	15–25 typical; 45–55 peak at month-end close and EOFY; idle out of hours
Availability	≥ 99.5% business-hours (Mon–Fri ~07:30–18:00); no 24/7 requirement
Recovery	RPO ≤ 1 hour (no loss of financial transactions); RTO ≤ 1 business day (≤ 8 business hours)
Data footprint	~22 GB SQL Server data, growing ~5 GB/year, plus scanned-document attachments
Retention	Financial records and audit logs retained ≥ 7 years
Licensing	Commercial — Ledgerline per-user + SQL Server; cloud licence treatment (licence-included vs BYOL) is material
Integrations	AD authentication; O365 SMTP; LMS fee-status; payroll-bureau SFTP; banking/payment-gateway file exchange
High availability	Out of baseline scope — Multi-AZ resilience deferred to the follow-on HA design

3. Review of Existing Architecture

Not applicable — this is a greenfield cloud foundation; there is no existing cloud baseline to review. The on-premises current state is documented separately in the Accounting System Server Specifications and the ICT Environment Overview.

4. Architecture Design

4.1 Assumptions and constraints

#	Assumption / constraint	Source
A1	Region must be ap-southeast-2 (Sydney) for financial-records data residency	Migration Requirements; Privacy Policy
A2	Application stack preserved: Windows Server 2016, Microsoft SQL Server, Ledgerline	Migration Requirements; Role Brief
A3	Data footprint ~22 GB SQL Server data, +~5 GB/year, plus scanned attachments	Application Spec; Server Specs
A4	Concurrent users 15–25 typical, 45–55 at month-end / EOFY; idle out of hours	Application Spec
A5	Financial records + staff/debtor PII must remain within Australia; retained ≥ 7 years	Privacy Act 1988 APP 8; financial-records obligations
A6	Preserve AD authentication over a private network to campus AD; staff-only access	Application Spec; Migration Requirements
A7	Baseline is NOT high-availability; Multi-AZ resilience deferred to the HA design	Scoping decision

4.2 AWS account and region

An AWS account scoped to the migration engagement, provisioned by YAT and shared with MTS for the build. Region ap-southeast-2 (Sydney); no deployment outside Australian regions, to satisfy financial-

records residency. Availability Zone ap-southeast-2a for the baseline; the follow-on HA design introduces a second AZ.

4.3 Identity and Access Management (IAM)

Group	Purpose	Indicative permissions
YAT-ICT-Admins	YAT ICT day-to-day ops post-handover	Read-only on infra; full CloudWatch/RDS/EC2 console; no IAM changes
MTS-Consultants	MTS during build + support	Full admin during build; reduced post-handover
Application-Service	EC2 instance role for Ledgerline	RDS read/write; S3 read/write (attachments); CloudWatch logs
Finance-Auditors	Finance / external auditors	Read-only on logs, metrics, configs (financial-audit support)

- MFA required for all YAT-ICT-Admins and MTS-Consultants users (Essential Eight; User Access Policy).
- No long-lived access keys for humans; programmatic access via IAM roles only; EC2 access via instance profile.
- Configuration decision left to the implementer: the MTS-Consultants permission boundary during build vs after handover.

4.4 Network topology

VPC 10.0.0.0/16 (room to expand), with DNS hostnames/resolution and VPC flow logs enabled. This is an internal, staff-only service — there is no public internet ingress to the application. Single-AZ subnets in ap-southeast-2a:

Subnet	CIDR	Tier	Internet-facing?
public-egress-a	10.0.1.0/24	NAT gateway only (outbound patching)	Egress only
private-app-a	10.0.11.0/24	Application / Ledgerline EC2 + internal ALB	No
private-data-a	10.0.21.0/24	Database (RDS for SQL Server)	No

- Internet Gateway used only for NAT egress (Windows Update, vendor patches); no inbound internet path to the app.
- Route tables: public-egress → IGW; private-app → NAT; private-data → no internet route.
- Staff reach the service over the campus Site-to-Site VPN; AD authentication runs over the same private link.
- The follow-on HA design adds the corresponding -b subnets in ap-southeast-2b.

4.5 Compute (EC2 + Auto Scaling)

- EC2: general-purpose x86 (e.g. m6i.large — final type a C1 implementer decision); Windows Server 2016 AMI; placed in private-app-a (no public IP).
- EBS: gp3 root 80 GB + a gp3 data volume sized by the implementer (footprint + 12-month growth + headroom).
- Auto Scaling Group: min 1 / desired 1 / max 2 (baseline); target-tracking on CPU at 70%; ELB+EC2 health checks; 300 s cooldown.

- The workload is business-hours and idle overnight; the follow-on HA design adds cross-AZ capacity for resilience (not for load).

4.6 Load balancing (ALB)

- Internal (private) ALB in private-app-a; HTTPS:443 listener forwarding to the Ledgerline target group — reachable by staff over the campus VPN, not from the internet.
- Target group = the ASG instances; HTTP health check on the application health endpoint (30 s; 2 unhealthy → out of service).
- TLS via an ACM-issued certificate for the internal Ledgerline DNS name (DNS strategy a C8 implementer decision).

4.7 Database (RDS for SQL Server)

- Amazon RDS for SQL Server, Standard edition (preserves the existing SQL Server engine and data); version confirmed against Ledgerline at build time.
- General-purpose instance class (e.g. db.m6i.large — a C2 implementer decision); gp3 storage sized to ~22 GB + ~5 GB/year growth.
- SQL Server licensing model — licence-included vs bring-your-own-licence — is a C3 implementer decision with a material cost impact.
- Multi-AZ DISABLED for the baseline (enabled in the HA design); storage encryption enabled (KMS).
- Placed in private-data-a; not publicly accessible. Automated backups + transaction-log backups sized to meet RPO ≤ 1 hour; backup window 22:00–04:00 AEST; maintenance Sun 02:00–06:00 AEST.
- Schema and data migration are YAT ICT's responsibility, not MTS's — MTS provisions an empty instance.

4.8 Storage

Resource	Type	Purpose
EC2 root volume	EBS gp3 80 GB	OS and Ledgerline application install
EC2 data volume	EBS gp3 (sized by implementer)	Ledgerline application data, report staging
Documents bucket	S3	Scanned invoices / purchase orders / supporting documents; lifecycle to Glacier for 7-year financial retention
Backups bucket	S3	Off-instance SQL backup exports, file snapshots

- Both buckets: block all public access; SSE-S3 (or SSE-KMS) encryption; versioning enabled; access logging to a log bucket; Object Lock considered for the 7-year financial-records hold.

4.9 Security

Security group	Inbound	Outbound
sg-alb	HTTPS:443 from the campus VPN / staff CIDR only	HTTPS to sg-app
sg-app	from sg-alb; RDP:3389 from MTS bastion (design left to implementer)	SQL:1433 to sg-db; HTTPS via NAT; LDAPS to campus AD; SMTP to 0365; SFTP to payroll bureau; banking endpoints
sg-db	SQL Server:1433 from sg-app only	none

- Encryption in transit: HTTPS staff→ALB→EC2; TLS EC2→RDS; AWS calls over TLS via VPC endpoints where available; LDAPS and SFTP for external links.
- Encryption at rest: EBS, RDS, and S3 all enabled (KMS where customer-managed keys are warranted for financial data).
- Operates under the AWS Shared Responsibility Model — AWS secures the cloud; YAT/MTS secure the OS, application, IAM, data and access in the cloud.

4.10 Monitoring (baseline)

Standard CloudWatch metrics for EC2, RDS, ALB and Auto Scaling. Baseline alarms (HA-tuned alarms come in the follow-on HA design):

Alarm	Threshold
EC2 CPU high	≥ 80% over 10 min
RDS CPU high	≥ 80% over 10 min
RDS free storage low	< 15%
ALB target 5XX / unhealthy host	> 5 / min or any unhealthy host
RDS connections high	> 80% of max connections

- Logging: VPC flow logs and RDS logs → CloudWatch Logs; ALB access logs → S3; EC2 OS logs via the CloudWatch Agent. Financial-audit-relevant logs retained to meet the 7-year obligation.

4.11 Naming and tagging conventions

Naming pattern yat-acct-<resource-type>-<env>-<az-or-purpose> (e.g. yat-acct-alb-prod). Mandatory tags:

Tag	Value
Project	YAT-Accounting-Migration
Environment	Production
Owner	YAT-ICT
ManagedBy	MTS-Migration during build → YAT-ICT post-handover
CostCentre	YAT-Accounting
DataClassification	Financial / Confidential (PII)

4.12 Backup

Resource	Mechanism	Retention
RDS database	Automated daily backups + transaction-log backups (point-in-time recovery)	Sized to RPO ≤ 1 h; long-term financial copies retained ≥ 7 years
EC2 EBS volumes	Daily AMI snapshot (Data Lifecycle Manager)	14 days
Documents (S3)	Versioning + lifecycle to Glacier	≥ 7 years (financial-records obligation)

- Cross-Region backup copies are out of scope for the baseline — addressed in the follow-on HA design.

4.13 Recovery objectives — baseline state

The baseline meets RPO $\leq$ 1 hour through RDS automated + transaction-log backups (point-in-time recovery), and supports RTO $\leq$ 1 business day via a single-AZ restore. The single AZ and the single RDS instance remain known single points of failure: tolerable for a business-hours service in the short term, but resilience against an AZ failure is the objective of the follow-on HA design.

4.14 Components requiring vertical scaling

RDS instance-class changes require a modify-and-apply (a brief interruption, taken in the maintenance window, outside business hours); EBS volumes support online resize with no downtime.

4.15 Single points of failure removed

Not applicable — this baseline is single-AZ by design; the single RDS instance and the single AZ are known single points of failure, deliberately deferred to the follow-on HA design.

4.16 Configuration decisions left to the implementer

The design is opinionated where it matters and silent where the implementer must show judgement. Each decision below is to be made and evidenced in the Deployment Report, justified against the Ledgerline workload.

#	Decision	Why left open
C1	EC2 instance type (general-purpose family)	Size against the Ledgerline workload (15–25 typical / 45–55 month-end peak)
C2	RDS for SQL Server instance class	Size against the SQL Server workload (read-heavy at month-end)
C3	SQL Server licensing model — licence-included vs BYOL	Material cost decision under the cloud operating model
C4	EBS data volume + RDS storage size	Compute from the ~22 GB footprint + ~5 GB/year growth
C5	Backup retention + RPO mechanism	Meet RPO $\leq$ 1 h and the 7-year financial-records retention
C6	Bastion / jump-host design for RDP admin access	Left to the implementer
C7	Security-group ingress specifics	Confirm AD/LDAPS, SMTP, payroll SFTP and banking endpoints
C8	DNS strategy + ACM certificate domain	Confirm the internal Ledgerline hostname with YAT ICT

5. Implementation Sequencing

Not applicable — this is a greenfield build in a new account, not a change to a running system; build order is at the implementer's discretion and there is no live service to sequence around or roll back.

6. Simulation and Verification Plan

Not applicable — failure/resize simulation and availability verification are the subject of the follow-on HA design and its deployment report; the baseline build is verified functionally per the Deployment Report's testing section.

7. Out of Scope

Stated explicitly so the implementer knows what not to build (these are the deliberate inputs to the follow-on HA design):

- Multi-AZ SQL Server; cross-AZ subnets (private-app-b, private-data-b); ASG capacity ≥ 2 across AZs.
- HA-tuned monitoring (cross-AZ latency, replica lag); cross-Region backup copies and DR runbook.
- Failure-simulation testing; automated availability reporting against the recovery objectives.
- And, out of MTS scope entirely: Ledgerline install, SQL Server data migration, cutover, and change management (YAT ICT).

8. References

- Accounting System Application Specification; Migration Requirements; Server Specifications; Operational Costing.
- Engagement Role Brief; Consultation Notes; ICT Environment Overview.
- Privacy Policy; User Access Policy; Security and Incident Response; Industry Standards Reference (AWS Well-Architected, ACSC Essential Eight).

Document control

Field	Value
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